

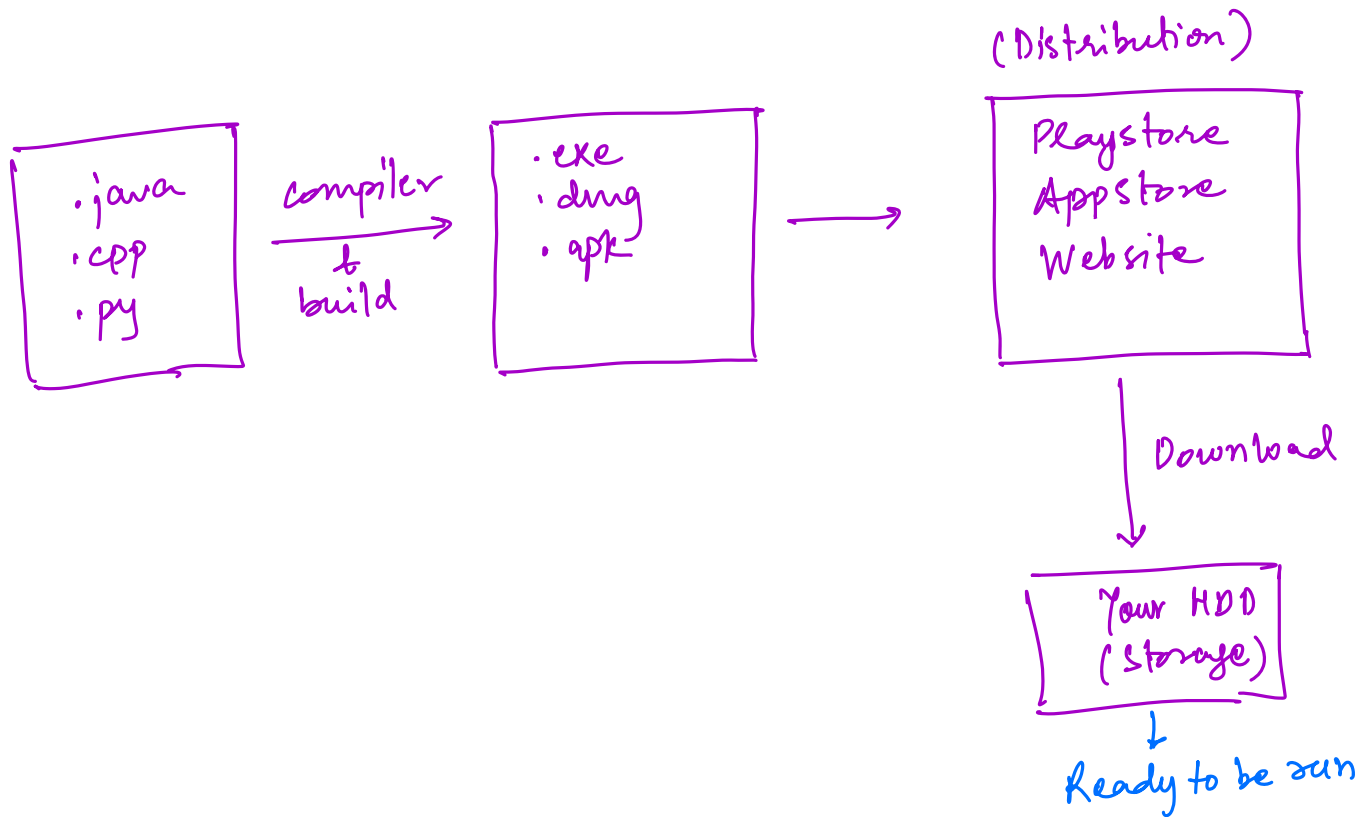
Agenda

- How a program executes?
- Threads
- Concurrency vs Parallelism
- Multicore Systems
- Steps to create a multithreaded program
- Write some multithreaded code.

→ How a program executes?

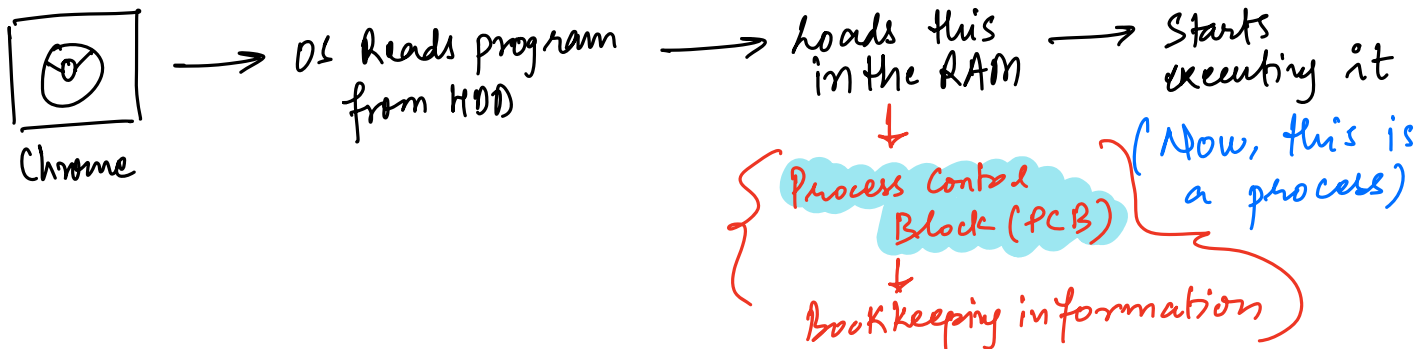
game, chrome, pdf scanner, etc.

.exe, .dmg, .apk, etc.



Program:- An application sitting on your storage (HDD/SSD) and is ready to be run (not running yet).

Process:- A program that is currently in execution.



→ CPU (Central Processing Unit)
2.5-4.0 GHz → perform billions of operations per second.

Mega → 10^6

Giga → 10^9

Tera → 10^{12}

Peta → 10^{15}

→ $\overline{a+b}$
→ Read the value of a } → RAM?
→ - - - - b } =
→ Add them together.

Reading 1MB from RAM → 0.25ms } → 80x slower
- - - 1MB - - HDD → 20ms.

Process Control Block → A data structure created and maintained by the OS which contains all the information about a process.

1 process → 1 PCB.

I ma Alok Singh, I m a Instrumtr

- Display input
 - Spell check
 - Grammar check
 - Auto-save
 - Formatting
- Simultaneously

Displaying char "I"
" "

Now i am spell checking ← WAIT
.....

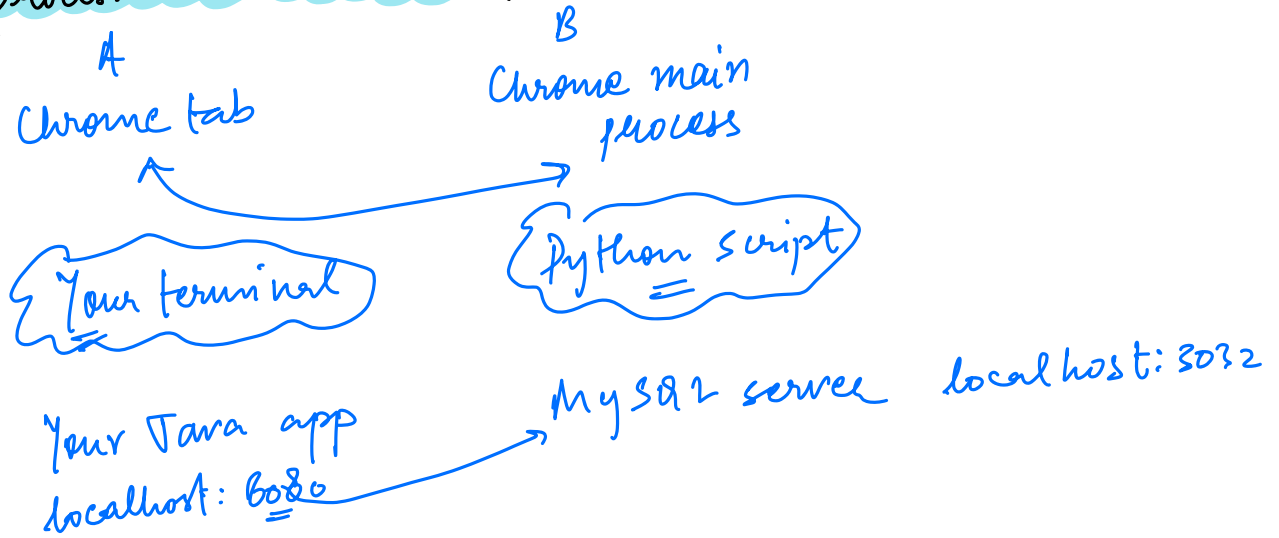
One process = One Program Counter = one task at a time

obvious → let's create 5 processes for 5 different tasks.

- Process 1 → Document Viewer →
- Process 2 → Spell checker →
- Process 3 → Grammar checker →
- ...

- Chrome (Bank account details)
- A game you downloaded

→ Interprocess Communication (IPC)

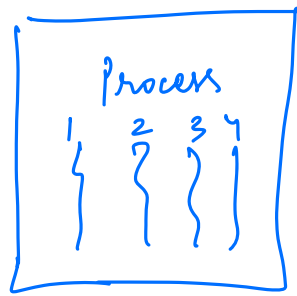


- Pipes
- Messaging Queues.
- Shared memory (OS supervision)
- Sockets.
- Files.

Create a new process → 10,000 - 100,000 μ s

Create a new thread. → 10 - 100 μ s. → 1000x faster.

- execute tasks in parallel
- can easily share data with other tasks.
- is lightweight to create
- lives within the same memory space.



→ Introduction to Threads.

CPU understands → instructions to execute.

CPU executes thread.

Your process always has a thread → main thread

	Process	Thread.
Memory	separate memory	share same memory
Communication	Complex (IPC)	Easy (direct memory access)
Creation time	slow (heavy)	fast
Isolation	Fully isolated	Share data
Crash impact	Does not affect other process	One crash can kill the whole process.

→ Single core v/s Multi-core systems.

concurrency without parallelism → single core CPU running multiple threads.
parallelism without concurrency → multiple cores running independent program. (main thread)
Both concurrency and parallelism → multiple threads and multiple cores.

P1 → T1, T2, T3,

P2 → T4, T5, T6

C1 T1, T5, T3

C2 T2, T1, T6

C3 T3, T6, T2